Virtual Mitosis Lab **Name** ______________________

Find the virtual lab at:
http://www.biology.arizona.edu/cell_bio/activities/cell_cycle/cell_cycle.html

Answer the questions below as you read the first two pages of the virtual lab website.

1. Why are root tips so useful for observing mitosis? ______________________

Root tips are useful because many cells are actively dividing there (high mitotic index).

2a. Describe what happens to the DNA in interphase. ______________________

During interphase, DNA is replicated in S phase to make sister chromatids.

______________________ 2b. Are the chromosomes visible? ______________________

No, chromosomes are not clearly visible in interphase.

3a. In what phase do the chromosomes become visible? ______________________

Prophase.

b. What happens to the nuclear membrane at this point? ______________________

The nuclear membrane breaks down.

4. During metaphase, why are the chromosomes organized to line up in the middle of the cell?

They line up at the metaphase plate so each daughter cell gets one copy of each chromosome.

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5. What is the main characteristic of anaphase? ______________________

Sister chromatids separate and move to opposite poles.

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6a. What are two specific events that occur in telophase? ______________________

Two events in telophase: nuclear membranes reform and chromosomes decondense.
Cytokinesis also begins.

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➢ Click NEXT at the bottom of the page. You do not need to copy the data table onto separate paper, as it's included here.

➢ Now you will begin the lab. You will be presented with 36 onion root tip cells. Determine which phase each cell is in. (You will know if you are correct or not.) Place a tally mark in the appropriate column in the table below. You should have a total of 36 tally marks when you are finished.

Interphase	Prophase	Metaphase	Anaphase	Telophase

➢ Complete the data table below by following the instructions for each row.

Data Table

	Interphase	Prophase	Metaphase	Anaphase	Telophase	Totals
# of cells	20	10	3	2	1	36
% of cells	55.56%	27.78%	8.33%	5.56%	2.78%	100%
# of minutes	588.9	294.4	88.3	58.9	29.4	1060

You'll need a calculator to complete some of the rows in this table.

- To fill in the **"# of cells"** row, count the number of cells in each of the phases from the data table on page 1 of this lab and write the number in the appropriate column.

- To fill in the **"% of cells"** row, divide the number of cells in each phase by 36 and then multiply by 100. (We use 36 because that's the total number of cells you observed.

- To fill in the **"# of minutes"** row you'll determine how long a cell spends in each phase. To to this, multiply the % of cells by 1060, which is the total number of minutes in an onion root tip cell cycle.

- You can check your work for each row by adding up the numbers in each column. The total should be exact in the # of cells row, and extremely close, if not exact, in the other two rows.

Analysis Questions

1. Which phase of the cell cycle had the most cells? ______________. Why do you think this is so? ______________

Interphase had the most cells because cells spend most of the cycle growing and copying DNA.

2. Describe how the cells in interphase looked compared to the cells in prophase.

Interphase cells: intact nucleus and diffuse chromatin.
Prophase cells: condensed visible chromosomes and nuclear membrane breakdown.

3. Describe how, the cells in metaphase looked compared to the cells in anaphase.

Metaphase cells have chromosomes lined up at the center; in anaphase chromatids separate apart.

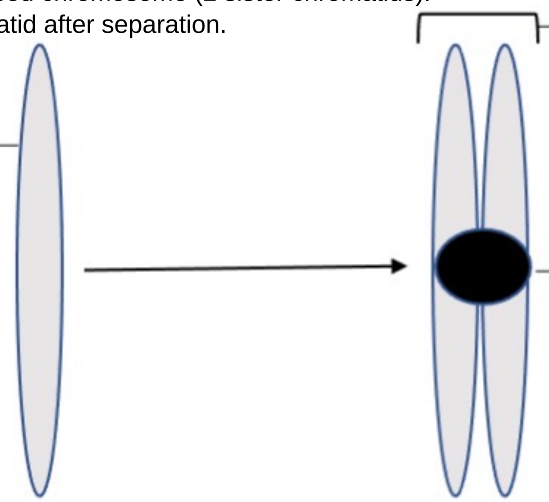
4. Are the daughter cells produced by mitosis identical to their parent cell or are they different? Explain your answer.

They are identical to the parent cell (except rare mutations) and keep the same chromosome number.

5. Label the illustration below using the following terms:

centromere, chromatid, sister chromatids, replicated chromosome, single chromosome

Centromere = pinched middle where chromatids attach.
Chromatid = one strand/half of a replicated chromosome.
Sister chromatids = identical pair joined at centromere.
Replicated chromosome = X-shaped chromosome (2 sister chromatids).
Single chromosome = one chromatid after separation.



6. What process causes the single chromosome to copy itself?

DNA replication during S phase of interphase.

7. How many times do the chromosomes replicate in mitosis?

Once.

8. How many daughter cells are produced during mitosis?

Two daughter cells.

9. What is the term used to describe the "pinching in" of the cell membrane during telophase?

Cytokinesis (formation of a cleavage furrow in animal cells).

10. In what phase do the chromosomes replicate?

Interphase (specifically S phase).

Label the phase each cell represents. Then describe the characteristics of each cell that helped you decide.

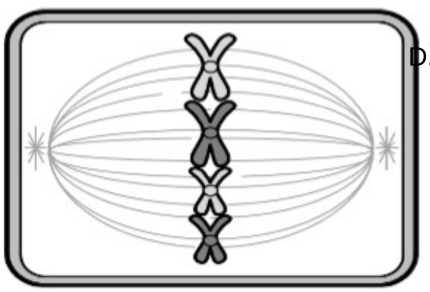
A. Prophase - chromosomes condense; nucleus starts breaking down.

B. Interphase - intact nucleus and uncondensed chromatin; no lined-up chromosomes.

C. Telophase/Cytokinesis - two nuclei reform and the cell pinches in.

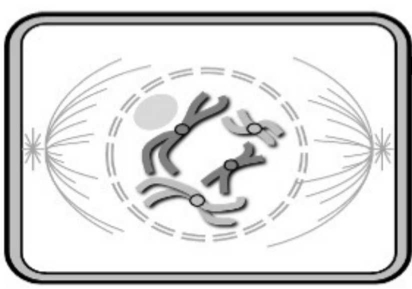
D.

D. Metaphase - chromosomes line up at the middle (metaphase plate).



E.

E. Prophase - chromosomes are visible and spindle fibers form.



F.

F. Telophase - chromosomes decondense and nuclei re-form in daughter cells.